D. Interactive Dashboard Architecture & Analytical User Guide

How to navigate and interpret the 5 interconnected exploratory modules in dashboard.html

Tab 1: Global Macroevolutionary Figures

`[ Fig 1A ]` `[ Fig 1B ]` `[ Fig 2 ]` `[ Fig 3 ]`

The Big-Picture Evolutionary Synthesis

- Fig 1A: Curated benchmark summary across 33 representative species.
- Fig 1B: Exhaustive distribution across all 68 vertebrate genomes.
- Fig 2: Mechanistic drivers (Bivariate GC regressions, gene length compaction, telomere distance).
- Fig 3: Time-calibrated phylogenetic chronogram, chromosome synteny maps, and codon ΔGC3 divergence.

Tab 2: 1-on-1 Head-to-Head Comparison

`[ Linear Tracks ]` `[ Inset Map ]` `[ Diagnostic Suite ]`

Direct Pairwise Syntenic & Structural Contrast

- Select Species A & B from searchable comboboxes (e.g., Human vs. Spotted Gar).
- Synchronized views: Linear Chromosome tracks, Bivariate with Mini Inset, or 6-Panel Diagnostic suites.
- Instant differential readout: ΔCDS GC%, ΔWhole-Gene GC%, and chromosomal context.

Tab 3: Single Species Deep Dive

`[ Interval Bounds ]` `[ Flanking Markers ]` `[ Ideograms ]`

Locus Anatomy & Architectural Metadata

- Comprehensive profile of any individual species in the 68-genome database.
- Exact genomic interval bounds, flanking syntenic markers, and gene count.
- High-resolution chromosome ideograms displaying exact physical coordinates.

Tab 4 & 5: Interval Database & Statistical Contrasts

`[ 1,530 Gene Models ]` `[ Wilcoxon Tests ]` `[ Effect Sizes ]`

Curated Gene Models & Hypothesis Testing

- Tab 4 (Database): Filterable, searchable table of all 1,530 curated gene models with z-scores.
- Tab 5 (Statistics): Two-sided Wilcoxon rank-sum tests, Hodges-Lehmann effect sizes, and FDR q-values.

Recommended Exploratory Workflow & Keyboard Shortcuts:

- Step 1 (Tab 1): Inspect macroevolutionary distributions➔ Step 2 (Tab 2): 1-on-1 syntenic contrasts➔ Step 3 (Tab 3): Deep dive interval architecture
- **Interactive Lightbox: Click any image to open fullscreen • Shortcuts: [Esc] Close • [0] Fit to screen • [+] / [-] Zoom • [1]–[6] Diagnostic panels**